

Exploring the Link Between Alcohol Consumption Frequency and Depression: Cross-Sectional
Study with Behavioral, Social and Demographic Determinants

Jueshen Hou

Contents

Abstract	3
Introduction.....	4
Study Methods & Measures	5
Table of Variables/Levels	7
Statistical Methods	8
Results.....	9
Table 1. Association Between Alcohol Consumption Frequency and Confounders.	12
Table 2. Bivariate Association Between Depression Frequency and Covariates.....	16
Table 3. Analysis of Maximum Likelihood Estimates and Odds Ratio of Model (include Employment and Income Variable Only Non-Missing Rows).....	21
Figure 1. Odds Ratio of Association Between Alcohol Consumption Frequency and Depression.....	23
Discussion.....	23
Reference.....	27
Appendix	29

Abstract

Depression is a mental health disorder that is influenced by multiple behavioral, social and demographical determinants, and alcohol consumption is often considered to be associated with depression severity. However, the association after considering variables across different arena of determinants was understudied. This study investigated the association between severity of depression and alcohol consumption frequency among U.S. adults aged between 18 and 85 after adjusting for multiple social, behavioral and demographic factors.

The study utilized cross-sectional design with data from National Health Interview Survey (2010-2019) and obtained through the IPUMS Health Surveys platform (Blewett et al., 2024). Binary logistic regression model was employed to assess association between depression severity (derived from depression frequency variable due to the absence of clinically standardized PHQ-9 depression severity measure in earlier years) and alcohol consumption frequency. Confounders including sex, age, race/ethnicity, hours of sleep, socioeconomic status, health insurance status and employment status were included in the model to allow holistic investigation of the association. Sensitivity analysis was conducted to examine the impact employment status and socioeconomic status (SES) and employment-related missingness.

A checkmark-shaped association was found between depression severity and alcohol consumption frequency. Participants with moderate alcohol consumption frequency had lower odds of severe depression, while daily and frequent drinkers exhibited elevated odds. An “Unknown” category was presented in alcohol consumption frequency with lowest odds ratio but yielded limited interpretability due to diverse composition of this category and uncovered underlying characteristics. Inclusion and exclusion of missing observations in SES and

employment status exhibited minimal difference on association trend, suggesting confounding effect, while excluding the two variable weekend association patterns.

Findings in this study aligned with prior research on the topic. Limitations include observational study design, lack of clinical depression severity measure, and information loss due to variable recoding. Future studies with longitudinal study design is needed for casual inference of the relationship.

Introduction

Depression is a mental health disorder with persistent feelings of sadness and lack of interest in daily activities, which impact the well-being across different age groups and populations (National Institute of Mental Health, 2024). Those who suffer from depression could have compromised quality of life due to the disorder, as well as undermined physical health conditions, which could eventually lead to decline of people's satisfaction with their quality of life (Cho et al., 2019). As one of the most prevalent mental health conditions, depression yield a profound burden to individuals who suffers from the disorder, as well as the society as a whole (Monroe & Harkness, 2022). Recent data from CDC reported that 18.5% of the U.S adults have been diagnosis with depression, with state model-based age-standardized prevalence of depression ranged from 12.7% to 27.5% and median at 19.9%. Thus, it is important for the field of public health to address and mitigate the impact of depression to a proper extent.

Research have shown that depression is a complex disorder influenced by various factors, including biological, behavioral social and demographic determinants (Remes et al., 2021). The consumption of alcoholic beverages, along with the potential abusive usage of alcohol, is a factor that is strongly associated with depression. Individuals with depression, or depressive symptoms

might see alcohol consumption as a resort to mitigate the symptoms they suffer from such disorder (Qi et al., 2024). A recent study from Tu, et al revealed an inverted U-shaped association between sleep disturbance and severity of depression, highlighted the importance of considering sleep quality when assessing severity of depression (Tu et al., 2024). Social and Demographic factors, such as biological gender and access to health care, can yield substantial influence to the severity of depression (Kuehner, 2017; Wang & Pu, 2024). Given the complex interplay nature between depression with various social and demographical factors, as well as its profound impact to the mental health and wellbeing of public across different groups, a comprehensive analysis that incorporating these interplayed factors is deemed to be necessary. While previous research has explored the association between individual factor or specific group of factors' association with severity of depression, holistic investigations that examine the association between alcohol consumption and depression severity after adjusting for a range of factors across social, demographical and behavior determinants is relatively understudied. To address this gap, the goal of this cross-sectional study is to examine the impact of alcohol consumption frequency toward severity level of depression after considering, health insurance status, family income, sleep quality, gender and employment status. This study builds on author's previous final project *"Association of Severity of Depressive Symptom and Weekly Alcohol Consumption Frequency after Considering Health Insurance, Sleep Hours and Smoking Status"* completed for [BIST 0615: Applied Categorical Data Analysis, Fall 2024], with significant modifications (Hou, 2024).

Study Methods & Measures

We conducted cross-sectional, observational study utilizing the National Health Interview Survey (NHIS) data set from 2010 to 2019. NHIS is a health survey program conducted annually by CDC to monitor the health of U.S. population across different geographic and demographic

groups. The survey was conducted toward non-institutionalized U.S. civilians through personal interview along with questionnaire (CDC,2024). 164532 participants with age range between 18 and 85 years old were included in the study sample. Our primary outcome variables consist of the frequency of experiencing depression symptoms (i.e. How often do you feel depressed), which consists of 9 levels: NIU (not in universe), Daily, Weekly, Monthly, A few times a year, Never, Unknown-refused, Unknown-not ascertained, Unknown-don't know. Levels including Unknown-refused, Unknown-not ascertained, Unknown-don't know and NIU in outcome variable are considered as missing and excluded for the modelling and bivariate association exploration, such observations with missing values were due to the survey design between 2010 to 2017, when only one quarter of the sample adults was asked for the relevant question. Although the clinical standard variable of depression severity based on Patient Health Questionnaire (PHQ_CAT9) was available for 2019, such variable was not available from 2010 to 2018. Thus, depression symptom frequency was selected as the alternative indicator of severity of depression. This variable collapsed into a binary variable for regression model building due to the violation of required statistical assumption, the details are available in statistical method section.

Alcohol consumption frequency (Frequency drank alcohol in past year: Days in past year) is considered as the primary predictor of interest, which ranges from 0 days to 365 days. Alcohol consumption frequency was recoded as a categorical variable with 6 levels: Non-drinker (0 days), Sometime drinker (1–80 days), Moderate drinker (81–150 days), Frequent drinker (151–312 days), and Daily drinker (313–365 days). Such categorization was designed to improve the interpretability through allowing comparisons between groups instead of tracking changes on a continuous basis, which might be difficult to interpret.

Seven confounders have been included in the analysis: age, sex, race, employment status, health insurance status, and hours of sleep. We decided to convert age into a categorical variable with 8 levels: 18-29, 30-39, 40-49, 50-59, 60-69, 70-79, 80-85 and “Missing” for interpretability, same approach was also performed on hours of sleep, which was categorized into the following 3 level: Short (1-5 hours), Normal (6-8 hours) and Long (8 hours and above). Sex, Race, Employment Status are categorical variables but coded as numeric in original form, new categorical version of these variables was separately created and utilized in the analysis. The rest of the variables were left unchanged. A comprehensive summary of variables and levels of variables can be found in the table below.

Table of Variables/Levels

Variable name	Description	Recode and Levels
Depression Frequency (DEPFEQ)	Frequency of experiencing depression symptoms	NIU(not in universe), Daily, Weekly, Monthly, A few times a year, Never, Unknown-refused, Unknown-not ascertained , Unknown-don't know.
Depression Frequency Binary (DEPFEQ_binary)	Frequency of experiencing depression symptoms	Severe (Daily, Weekly, Monthly) Not Severe (A few times a year, Never)
Alcohol Consumption Frequency (ALCDAYSyr)	Frequency drank alcohol in past year: Days in past year	Continuous (1-365)
Alcohol Consumption Frequency in categorical form (ALCDAYSyr_cate)	Frequency drank alcohol in past year: Days in past year	Non-drinker, Sometime drinker, Moderate drinker, Frequent drinker, Daily drinker
Age (Age)	Age	Continuous (18-85)
Age group(Age_Cate)	Age group people belong	18-29, 30-39, 40-49, 50-59, 60-69, 70-79, 80-85
Sex (SEX_cate)	Biological sex	Male and Female
Race (RACENEW_cate)	Self-reported Race (Post-1997 OMB standards)	White Only, African American, Asian, American Indian/Alaska Native, Multiple Races, Other Races

Employment status(EMPSTATIMP5_cate)	Employment status	Employed, Unemployed
Health insurance status (HINOTCOVE_cate)	Health Insurance coverage status	No, has coverage Yes, has no coverage
Hours of sleep (hrsleep)	Usual hours sleep per day	Continuous (1-24)
Hours of sleep in categorical form (hrsleep_cate)	Usual hours sleep per day (categorical)	Short (1-5 hours) Normal (6-8 hours) Long (8+ hours)
Total Family Income (INCFAM97ON2_cate)	Total combined annual income of a family household, an indicator of socioeconomic status	\$0- \$34,999 \$35,000 – \$74,999 \$75,000 - \$99,999 \$100,000 and over

Statistical Methods

We applied multiple statistical models to conduct data analysis, and generated table of sample characteristics (table 1) and bivariate association (table 2), as well as logistic regression model to explore the association between severity of depression and alcohol consumption frequency after considering a series of behavior, social and demographical confounders. All the analyses was conducted utilizing SAS Enterprise Guide (Version 8.4), and accounted for the primary sampling unit, cluster, and sampling weight as outlined in the NHIS Procedures. Table 1 includes the frequency and row percentage of levels in predictor of interest and confounders. Rao-Scott Chi-square test was utilized to assess the association between cofounder and primary predictor of interest. Table 2 contains the assessment of bivariate associations between response variable and predictor / confounders. We consider associations with p-values less than 0.05 as statistically significant associations for table 1 and 2.

An exploratory model with depression frequency and alcohol consumption frequency was created to test the assumptions required for proportional odds model via score test and violation of required assumption was determined. Therefore, the outcome variable was recoded to a binary

variable with Severe (Daily, Weekly and Monthly) and Not Severe (A few times a year, Never) . This binary design retained meaningful information of outcome variable and helps to improve interpretability and avoid violation of proportional odds assumption. Adjacent odds model was not utilized due to technical limitations of SAS Enterprise Guide and its limited interpretability. Survey-adjusted statistical methods were applied in the analysis to accommodate the complex survey design of NHIS data through incorporating sample weight, strata and primary sample units (PSU) in the analysis. The final model was constructed between the binary severity variable and alcohol consumption frequency, along with the 7 confounders.

Three models were fit to assess the robustness of the association and the impact of missing category in two variables with the highest proportion of missing values (Employment Status and Total Family Income). Model 1 was based on full data set and included primary predictor of interest with confounders but excluding employment status and total family income (which have 18.92% and 25.15% of missing value, respectively), model 2 included all the confounders but was fitted based on observations without missing value in the two variables. Model 3 included all the confounders and was fitted on the full dataset, and missing values were coded as a separate category. This design allowed us to determine whether the missingness in these two variables have meaningful impact to the association. The missingness in other confounders was minimal, with the highest missing rate been 1.09%, therefore the inclusion of them was deemed appropriate for this research.

Results

The study sample includes 164523 people residing in the USA who are aged between 18 years old and 85 years old across different race and demographics. 6588 participants (3.76%) were

reported as daily drinkers (313-364 days per year), 10827 (6.59%) were reported as frequent drinkers (151-321 days), 10633 (6.64%) were reported as moderate drinkers (81-150 days), 57140 (36.22%) were reported as sometime drinker (1-180 days), 20444 (11.6%) were reported as non-drinkers, and 58630 (35.74%) were reported with unknown drinking status. Significant association was reported between health insurance coverage status and drinking frequency ($p < 0.0001$). The majority of participants across the alcohol consumption frequency groups reported health insurance coverage. 90.39% daily drinkers, 90.15% of the frequent drinkers, 88.75% of moderate drinkers, 86.46% sometime drinkers, 88.14% non-drinkers and 86.24% status unknown drinkers were reported with coverage. Strong association was also observed between employment status and alcohol consumption frequency ($p < 0.00001$), elevated percentage of employed participants were reported in frequent drinkers (78.08%), moderate drinkers (80.99%) and sometime drinkers (74.82%). Additionally, statistically significant association between sex and alcohol consumption frequency was reported in the analysis ($p < 0.0001$). 65.69% of the daily drinkers and 61.56% of the frequent drinkers were reported to be males, whereas only 34.31% of daily drinkers and 38.44% of frequent drinkers were reported to be females, more than half of the sometime drinkers and non-drinkers were reported to be females. Race, age, hours of sleep and socioeconomic status (depicted by total family income) also presented statistically significant association with the alcohol consumption frequency.

A detailed distribution of confounders across alcohol consumption categories are presented in Table 1. P values for association with alcohol consumption frequency were calculated through Rao-Scott Chi Square Test. All percentages in the table are row percentages (percentages against total population in each of the alcohol consumption categories), the counts of participants in each

categories were unweighted counts, whereas the their percentages were weighted percentages that are considered representative to the U.S. population.

Table 1. Association Between Alcohol Consumption Frequency and Confounders.

[illegible]

Characteristic	Level	Overall	Days drank alcohol past year: Daily Drinker (313-365 days)	Days drank alcohol past year: Frequent Drinker (151-312 days)	Days drank alcohol past year: Moderate Drinker (81-150 days)	Days drank alcohol past year: Sometime Drinker (1-80 days)	Days drank alcohol past year: Non Drinker	Days drank alcohol past year: Unknown	P-value*
Overall	--	164532 (100.00%)	6588 (3.76%)	10827 (6.59%)	10633 (6.64%)	57410 (36.22%)	20444 (11.06%)	58630 (35.74%)	--
Race	American Indian/Alaska Native	1996 (1.14%)	37 (0.49%)	78 (0.59%)	87 (0.75%)	581 (0.91%)	333 (1.52%)	880 (1.49%)	
	Asian only	9246 (5.76%)	159 (2.45%)	325 (2.89%)	346 (2.99%)	3105 (5.34%)	740 (3.73%)	4571 (8.21%)	
	Black/African American only	21227 (12.09%)	512 (6.76%)	837 (6.94%)	1066 (8.98%)	7192 (11.35%)	2954 (12.91%)	8666 (14.67%)	
	Missing	1801 (1.41%)	3 (0.04%)	11 (0.10%)	17 (0.14%)	133 (0.19%)	57 (0.26%)	1580 (3.64%)	
	Multiple Race (1999-2018: Incl	2679 (1.50%)	101 (1.33%)	238 (1.84%)	214 (1.82%)	1250 (2.07%)	438 (1.95%)	438 (0.67%)	
	Other Race	360 (0.25%)	0	0	0	0	0	360 (0.69%)	
	White only	127223 (77.85%)	5776 (88.92%)	9338 (87.64%)	8903 (85.32%)	45149 (80.14%)	15922 (79.63%)	42135 (70.63%)	
SES	\$0 - \$34,999	48901 (22.39%)	1823 (20.27%)	2603 (17.45%)	2825 (19.16%)	18625 (24.12%)	9673 (37.70%)	13352 (17.63%)	<.0001
	\$35,000 - \$74,999	37547 (22.87%)	1903 (27.80%)	2969 (24.85%)	2942 (25.65%)	17249 (29.10%)	5599 (29.96%)	6885 (12.96%)	
	\$75,000 - \$99,999	13653 (9.50%)	804 (13.28%)	1340 (13.32%)	1365 (13.20%)	6635 (12.96%)	1570 (9.51%)	1939 (4.22%)	
	\$100,000 and over	24234 (20.09%)	1640 (32.00%)	3377 (38.92%)	2905 (35.87%)	11350 (26.79%)	2060 (14.10%)	2902 (7.50%)	
	Missing	40197 (25.15%)	418 (6.65%)	538 (5.47%)	596 (6.13%)	3551 (7.03%)	1542 (8.73%)	33552 (57.69%)	

(continues on next page)

Characteristic	Level	Overall	Days drank alcohol past year: Daily Drinker (313-365 days)	Days drank alcohol past year: Frequent Drinker (151-312 days)	Days drank alcohol past year: Moderate Drinker (81-150 days)	Days drank alcohol past year: Sometime Drinker (1-80 days)	Days drank alcohol past year: Non Drinker	Days drank alcohol past year: Unknown	P-value*
Overall	--	164532 (100.00%)	6588 (3.76%)	10827 (6.59%)	10633 (6.64%)	57410 (36.22%)	20444 (11.06%)	58630 (35.74%)	--
Age	18-29	27452 (21.29%)	360 (6.26%)	1827 (17.93%)	2301 (23.59%)	11506 (23.08%)	1285 (8.35%)	10173 (25.27%)	<.0001
	30-39	27301 (17.08%)	608 (9.75%)	1971 (18.94%)	2272 (21.56%)	11456 (19.98%)	2029 (11.19%)	8965 (15.57%)	
	40-49	25513 (16.82%)	816 (14.14%)	1842 (18.53%)	1836 (18.01%)	10302 (19.14%)	2556 (14.42%)	8161 (14.96%)	
	50-59	27943 (17.19%)	1377 (22.31%)	2075 (20.05%)	1893 (18.16%)	9798 (17.19%)	3988 (20.40%)	8812 (14.95%)	
	60-69	27803 (14.69%)	1665 (24.55%)	1861 (15.09%)	1462 (12.41%)	8105 (12.35%)	4772 (21.60%)	9938 (14.23%)	
	70-79	17853 (8.35%)	1167 (15.53%)	914 (6.90%)	651 (4.82%)	4201 (5.73%)	3550 (15.02%)	7370 (9.11%)	
	80-85	10605 (4.55%)	595 (7.45%)	337 (2.56%)	218 (1.46%)	2042 (2.54%)	2264 (9.02%)	5149 (5.83%)	
	Missing	62 (0.03%)	0	0	0	0	0	62 (0.10%)	
Sleep	Long	43057 (25.95%)	742 (10.26%)	674 (5.81%)	631 (5.63%)	4167 (7.02%)	2615 (12.38%)	34228 (58.48%)	<.0001
	Normal	108770 (66.58%)	5313 (82.10%)	9340 (86.89%)	9193 (86.66%)	47630 (83.48%)	15435 (76.13%)	21859 (37.39%)	
	Short	12705 (7.47%)	533 (7.63%)	813 (7.31%)	809 (7.71%)	5613 (9.50%)	2394 (11.49%)	2543 (4.13%)	

Bivariate analysis conducted between frequency of depression and covariates demonstrated significant association between alcohol consumption frequency and depression frequency ($p < 0.0001$). Participants who never felt depressed tend to have higher rates of sometime drinkers (35.81%) and status unknown drinkers (36.91%) compared to other drinking frequency categories. Notably, non-drinkers were reported to have higher prevalence of daily depression symptoms compare to other alcohol consumption frequency categories. Statistically significant associates were found between depression frequency and health insurance status, age, sex , socioeconomic status (Total Family Income), employment status and hours of sleep, respectively. Participants without health insurance are more likely to report suffering from daily depression (17.24%) when compared with those who lack insurance and never felt depressed (11.98%). Younger (18-29) and middle age group (50-59) was found to have higher rate of having higher frequency of depression feeling compared to other age groups. It was found that participants are individuals with lower socioeconomic status, and participants who have short duration of sleep tend to suffer from more frequent depression symptoms. A more detailed bivariate association for depression frequency and confounder can be found in Table 2. All percentages in the table are weighted percentages against total population in each depression severity category, whereas the corresponding counts are unweighted counts of population in each category.

Table 2. Bivariate Association Between Depression Frequency and Covariates

Predictor	Overall**	How often feel depressed: Daily	How often feel depressed: Weekly	How often feel depressed: Monthly	How often feel depressed: A Few Times A Year	How often feel depressed: Never	P-value*
***	164532	7155 (3.87%)	8820 (5.18%)	10087 (5.99%)	40717 (24.08 %)	97753 (60.88%)	
Primary Predictor: Frequency drank alcohol in past year							<0.0001
Daily drinker	6588 (3.76%)	298 (3.73%)	369 (3.85%)	400 (3.72%)	1566 (3.70%)	3955 (3.78%)	--
Frequent drinker	10827 (6.59%)	294 (4.45%)	600 (6.61%)	794 (7.89%)	3054 (7.48%)	6085 (6.25%)	--
Moderate drinker	10633 (6.64%)	270 (4.05%)	501 (5.93%)	616 (5.95%)	2793 (6.95%)	6453 (6.81%)	
Sometime Drinker	57410 (36.22%)	2177 (31.55%)	3044 (35.62%)	3725 (37.37%)	14866 (37.84%)	33598 (35.81%)	
Non-drinker	20444 (11.06%)	1487 (19.18%)	1383 (13.84%)	1214 (11.29%)	4804 (10.63%)	11556 (10.45%)	
Unknown	58630 (35.74%)	2629 (37.04%)	2923 (34.16%)	3338 (33.77%)	13634 (33.40%)	36106 (36.91%)	
Health Insurance Status							<0.0001
Yes	144254 (87.11%)	6068 (82.44%)	7533 (85.19%)	8668 (85.32%)	35772 (87.37%)	86213 (87.64%)	
No	19782 (12.48%)	1069 (17.24%)	1266 (14.48%)	1382 (14.11%)	4818 (12.13%)	11247 (11.98%)	
Missing	496 (0.41%)	18 (0.33%)	21 (0.33%)	37 (0.57%)	127 (0.50%)	293 (0.38%)	

(continues next page)

Predictor	Overall**	How often feel depressed: Daily	How often feel depressed: Weekly	How often feel depressed: Monthly	How often feel depressed: A Few Times A Year	How often feel depressed: Never	P-value*
***	164532	7155 (3.87%)	8820 (5.18%)	10087 (5.99%)	40717 (24.08 %)	97753 (60.88%)	
Age							0.0002
18-29	27452 (21.29%)	799 (15.81%)	1565 (23.80%)	2190 (28.04%)	6616 (20.42%)	16282 (21.11%)	
30-39	27301 (17.08%)	919 (13.55%)	1404 (16.31%)	1871 (17.87%)	7181 (17.82%)	15926 (17.00%)	
40-49	25513 (16.82%)	1220 (18.43%)	1383 (16.58%)	1655 (17.31%)	6453 (17.15%)	14802 (16.56%)	
50-59	27943 (17.19%)	1788 (23.67%)	1729 (18.55%)	1660 (15.53%)	7000 (17.80%)	15766 (16.58%)	
60-69	27803 (14.69%)	1345 (16.20%)	1459 (13.95%)	1437 (11.93%)	6919 (14.84%)	16643 (14.87%)	
70-79	17853 (8.35%)	650 (7.39%)	778 (6.76%)	820 (6.32%)	4143 (7.76%)	11462 (8.98%)	
80-85	10605 (4.55%)	431 (4.89%)	500 (4.03%)	453 (2.98%)	2400 (4.20%)	6821 (4.86%)	
Missing	62 (0.03%)	3 (0.05%)	2 (0.03%)	1 (0.02%)	5 (0.01%)	51 (0.04%)	
Employment Status							<0.0001
Employed	84203 (54.07%)	2140 (30.59%)	2140 (30.59%)	5029 (51.81%)	21138 (54.20%)	52221 (56.57%)	
Unemployed	49299 (27.01%)	3643 (49.00%)	3643 (49.00%)	2953 (26.69%)	11145 (25.27%)	28113 (25.56%)	
Missing	31030 (18.92%)	1372 (20.41%)	1372 (20.41%)	2105 (21.50%)	8434 (20.53%)	17419 (17.87%)	
Hours of Sleep							<0.0001
Long	43057 (25.95%)	2360 (34.10%)	2578 (29.42%)	2908 (29.26%)	11170 (27.15%)	24041 (24.34%)	
Normal	108770 (66.58%)	3230 (44.76%)	4982 (56.40%)	6131 (60.60%)	26478 (65.33%)	67949 (69.92%)	

(continues next page)

Predictor	Overall**	How often feel depressed: Daily	How often feel depressed: Weekly	How often feel depressed: Monthly	How often feel depressed: A Few Times A Year	How often feel depressed: Never	P-value*
***	164532	7155 (3.87%)	8820 (5.18%)	10087 (5.99%)	40717 (24.08 %)	97753 (60.88%)	
Sex							<0.0001
Male	74436 (48.32%)	2685 (41.13%)	3382 (40.10%)	3681 (38.16%)	16368 (42.98%)	48320 (52.58%)	
Female	90093 (51.68%)	4469 (58.87%)	5438 (59.90%)	6406 (61.84%)	24349 (57.02%)	49431 (47.42%)	
Missing	3 (0.00%)	1 (0.01%)	0	0	0	2 (0.01%)	
SES							<0.0001
\$0 - \$34,999	48901 (22.39%)	3740 (43.75%)	3740 (33.07%)	3553 (26.92%)	11810 (22.10%)	26058 (19.79%)	
\$35,000 - \$74,999	24234 (20.09%)	1194 (19.97%)	1847 (23.01%)	2252 (23.18%)	9343 (22.93%)	22911 (22.99%)	
\$75,000 - \$99,999	37547 (22.87%)	267 (5.02%)	491 (7.32%)	737 (8.95%)	3407 (9.33%)	8751 (10.10%)	
\$100,000 and over	13653 (9.50%)	316 (6.26%)	703 (12.19%)	1101 (15.26%)	5919 (20.15%)	16195 (22.09%)	
Missing	40197 (25.15%)	1638 (25.01%)	2039 (24.41%)	2444 (25.69%)	10238 (25.49%)	23838 (25.03%)	
Short	12705 (7.47%)	1565 (21.14%)	1260 (14.18%)	1048 (10.14%)	3069 (7.52%)	5763 (5.74%)	

(continuous on next page)

Predictor	Overall**	How often feel depressed: Daily	How often feel depressed: Weekly	How often feel depressed: Monthly	How often feel depressed: A Few Times A Year	How often feel depressed: Never	P-value*
***	164532	7155 (3.87%)	8820 (5.18%)	10087 (5.99%)	40717 (24.08 %)	97753 (60.88%)	
Race							<.0001
American Indian/Alaska Native	1996 (1.14%)	134 (1.69%)	135 (1.38%)	118 (1.05%)	505 (1.12%)	1104 (1.10%)	
Asian only	9246 (5.76%)	207 (2.50%)	294 (3.50%)	384 (3.69%)	2139 (5.52%)	6222 (6.46%)	
Black/African American only	21227 (12.09%)	927 (11.76%)	1020 (10.96%)	1102 (10.03%)	4812 (10.83%)	13366 (12.91%)	
Missing	1801 (1.41%)	84 (1.61%)	72 (1.02%)	93 (1.20%)	426 (1.41%)	1126 (1.46%)	
Multiple Race (1999-2018: Incl	2679 (1.50%)	186 (2.10%)	222 (2.34%)	198 (1.97%)	714 (1.71%)	1359 (1.26%)	
Other Race	360 (0.25%)	17 (0.34%)	26 (0.30%)	34 (0.39%)	103 (0.29%)	180 (0.21%)	
White only	127223 (77.85%)	5600 (79.99%)	7051 (80.51%)	8158 (81.68%)	32018 (79.12%)	74396 (76.61%)	

Among the three models fitted, we found that model 1 (the model without employment status and socioeconomic status) lacks a clear trend of odds ratios of having severe depression across alcohol consumption frequency categories. Odds ratio trends from Model 2 (the model with employment and socioeconomic variable but only regressed on observations without missing in these two variables), and model 3 (the model based on full data set but treat missing as a separate category), showed minimal difference. Thus, model 2 was selected as final model for investigating the association. Odds ratio table of model 1 and model 3 can be found in the Table A.1 and Table A.2 in appendix.

It was found that daily drinkers [OR: 1.058 (95%CI: 0.959,1.168)] and frequent drinkers [OR: 1.066 (95%CI: 0.974,1.168)] indicated slightly higher odds of experiencing severe depression compared to non-drinkers, however these odd ratios were not statistically significant. On the other hand, moderate drinkers [OR: 0.796 (95%CI: 0.724, 0.875)] and sometime drinkers [OR: 0.849 (95%CI: 0.801, 0.899)] exhibited statistically significant lower odds of suffering from severe depression. Participants with unknown drinking frequency demonstrated noticeably lower odds ratio compared to non-drinkers [OR: 0.568 (95%CI: 0.529, 0.611)]. The odds ratios of each category of alcohol consumption frequency exhibited a checkmark shaped pattern, individuals with unknown drinking frequency have the lowest odds of experiencing severe depression symptoms. However, participants with moderate alcohol consumption frequency had the lowest odds ratio, excluding unknown category. The overall trend of odds ratios exhibited a checkmark-shaped form. The odds ratio trend plot of model 1 (Employment and SES absent) shows a flatten trend, with odds ratios less than 1 across all categories. Details about the odds ratio, corresponding regression estimates of the association, and plot of odds ratios can be found in the table and figure below, and Figure A.1 in appendix A.

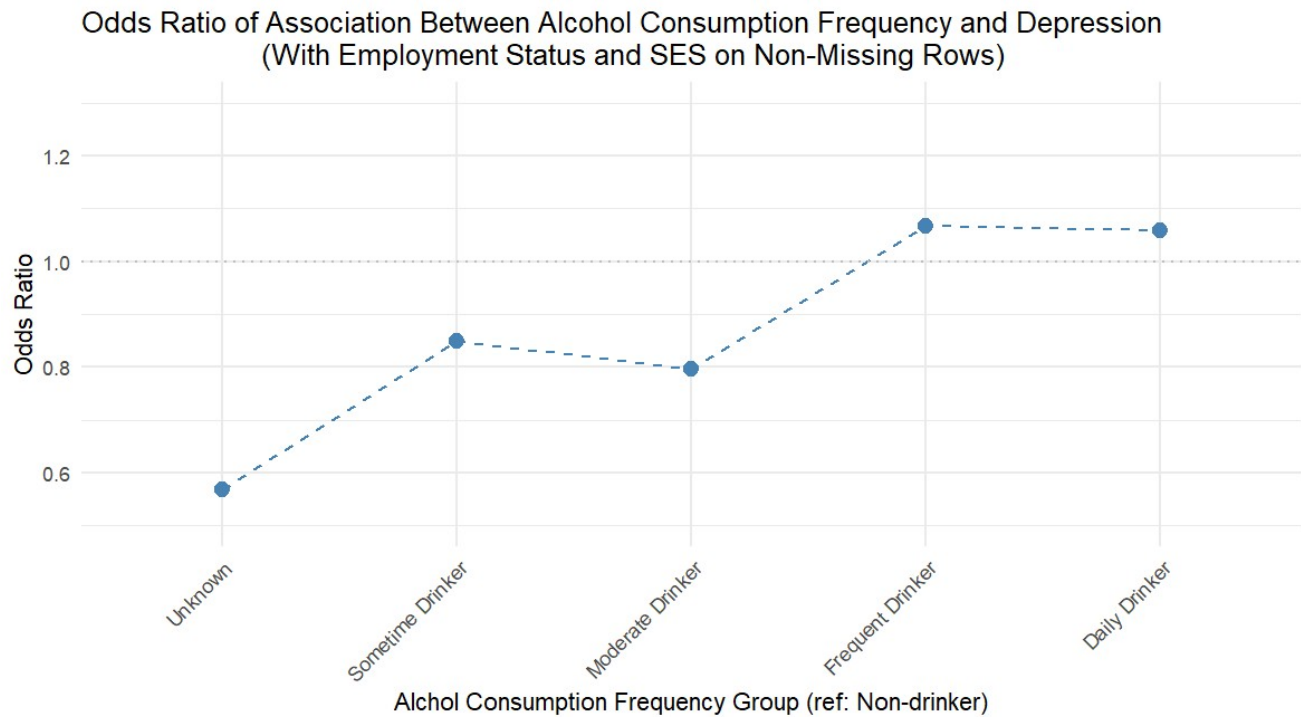
Table 3. Analysis of Maximum Likelihood Estimates and Odds Ratio of Model (include Employment and Income Variable Only Non-Missing Rows)

Table of Odds Ratios:

Variable	Category Comparison	OR	95% CI
Alcohol Consumption Frequency (ALCDAYSyr_cate)	Daily drinker vs Non-drinker	1.058	(0.959, 1.168)
	Frequent drinker vs Non-drinker	1.066	(0.974, 1.168)
	Moderate drinker vs Non-drinker	0.796	(0.724, 0.875)
	Sometime Drinker vs Non-drinker	0.849	(0.801, 0.899)
	Unknown vs Non-drinker	0.568	(0.529, 0.611)
Age Group (Age_cate)	30-39 vs 18-29	0.885	(0.889, 1.024)
	40-49 vs 18-29	1.000	(1.055, 1.229)
	50-59 vs 18-29	0.983	(1.000, 1.156)
	60-69 vs 18-29	0.683	(0.637, 0.744)
	70-79 vs 18-29	0.462	(0.391, 0.476)
	80-85 vs 18-29	0.422	(0.314, 0.397)
	Missing vs 18-29	0.679	(0.196, 0.974)
Sex (Sex_cate)	Female vs Male	1.499	(1.387, 1.517)
	Missing vs Male	2.287	(0.350, 37.022)
Race (Race_cate)	American Indian/Alaska Native vs White only	0.894	(0.506, 0.828)
	Asian only vs White only	0.545	(0.506, 0.636)
	Black/African American only vs White only	0.687	(0.610, 0.694)
	Missing vs White only	0.661	(0.636, 1.418)
	Multiple Race (1999-2018: Incl vs White only)	1.240	(1.053, 1.406)
	Other Race vs White Only	1.126	(0.837, 1.464)

Employment Status(EMPSTATIMP5_cate)	Employed vs Unemployed	0.554	(0.506, 0.562)
	Missing vs Unemployed	1.158	(1.055, 1.326)
Health Insurance Status (HINOTCOVE_cate)	Missing vs Yes, has no coverage	1.031	(0.603, 1.482)
	No, has coverage vs Yes, has no coverage	0.961	(0.954, 1.087)
SES(Total Family Income) INCFAM97ON2_cate	\$0 - \$34,999 vs \$75,000 - \$99,999	1.923	(1.841, 2.151)
	\$100,000 and over vs \$75,000 - \$99,999	0.747	(0.674, 0.803)
	\$35,000 - \$74,999 vs \$75,000 - \$99,999	1.251	(1.177, 1.373)
	Missing vs \$75,000 - \$99,999	0.882	(0.803, 0.980)
Hours of Sleep (HRSLEEP_cate)	Long vs Short	0.654	(0.628, 0.739)
	Normal vs Short	0.391	(0.382, 0.429)

Figure 1. Odds Ratio of Association Between Alcohol Consumption Frequency and Depression



Discussion

A checkmark shaped association between the alcohol consumption frequency and severity of depression was found in this study. Participants categorized as moderate drinkers on alcohol consumption frequency (81-150 days/year) showed the lower odds [OR: 0.796 (95%CI: 0.724, 0.875)] of experiencing severe depression compared to non-drinkers. Participants with unknown drinking frequency exhibited much lower odds ratio [OR: 0.568 (95%CI: 0.529, 0.611)] compared to people with moderate drinking frequency. However, the “Unknown” category consists of individuals who are either ineligible or reported unknown responses (not ascertained, refused and don’t know), and should be interpreted with caution. The result suggested that a light-to-moderate alcohol consumption frequency is associated with lower odds of experiencing severe depression.

The finding of our study aligns with prior research that suggested lower odds of severe depression among moderate drinkers. Studies completed from Visontay et al. have revealed a reduced risk of depression among people with moderate alcohol consumption frequency (Visontay et al., 2023). Association observed in the study also partially resembles the J-shaped association revealed in the study from Gémes et al. in 2019 and study from Kim in 2015 with behavioral and demographical factors, where non-drinkers and excessive drinkers were found to have elevated risk of suffering from severe depression, suggesting a complicated interplaying effect of such factors toward the association (Gémes et al., 2019; Kim et al., 2015). The check mark shaped trend observed in this study reiterated the importance of considering behavioral patterns and demographical characteristics in research on depression. Difference between the clarity of odds ratio trend between model 1 and model 2/3, as well as the below-1 odds ratio across all alcohol consumption frequency categories in model 1, demonstrated the importance of including social determinants, such as employment and socioeconomic status when examine the association between alcohol consumption frequency and depression, and suggested their potential confounding effect on the association (Assanangkornchai et al., 2020). Additionally, the minimal difference in odds ratio between model 2 and model 3 suggests that whether observations with missing in these two variables were treated as separate category did not significantly affect the association.

One notable unexpected result in this study was the unusually low odds of severe depression in the “unknown” alcohol consumption frequency group[OR: 0.568 (95%CI: 0.529, 0.611)], which consists of participant who are NIU (not in universe of corresponding survey question about alcohol consumption), and those who refused, were not ascertained, or don’t know about their alcohol consumption frequency. The unusual low odds ratio could suggest the presence of a

subpopulation with hidden characteristics that are associated with both alcohol consumption and depression, further research is needed for revealing details about this unique phenomenon, as well as its implication for the association (Torvik et al., 2011). Due to the consistency between the overall pattern and outcome of the association (particularly the decreased odds of severe depression in moderate alcohol consumption frequency category) and findings from previous studies, the observed trend in this study is considered valid and interpretable.

Our study has several limitations. Firstly, the observational study design and cross-sectional nature of NHIS data hindered the investigation of cause-effect relationships between alcohol consumption frequency and depression severity. Secondly, due to the absence of clinical depression severity variable, the self-reported depression frequency was chosen as alternative, which may lead to limited precision in the findings. Thirdly, the incident of unknown category in alcohol consumption frequency variable, could lead to potential bias in the examination of association. Additionally, depression severity was collapsed into a binary variable due to the limited functionality of statistical package utilized in this study, thus, information loss due to this alternative approach can't be ruled out.

Future studies should attempt to utilize longitudinal data to investigate the association between alcohol consumption frequency and depression severity so that casual inference could be performed to bring more robust outcome of the association. Utilizing standardized clinical assessment variable, such as PHQCAT (Patient Health Questionnaire depression scale) is also recommended, as it could help to enhance the validity. Stratified analysis based on social and demographic determinants is also recommended for uncovering modification effect or population specific trend of association.

In conclusion, this study determined a checkmark-shaped association between alcohol consumption frequency and severity of depression among U.S adults between 18 and 85 years old, where people with moderate alcohol consumption frequency exhibited the lowest odds of suffering from severe depression among participants with known alcohol consumption frequency category. The unknown category in alcohol consumption frequency showed the lowest odds ratio, however, such finding must be interpreted with caution as the unknown category consist of participants who were non-responsive on the corresponding survey question or not included due to criteria eligibility of survey questionnaire. The subgroup was retained for preserving the sample size and information, but future investigation is needed for uncovering the underlying characteristics of this group. While limitations induced by cross-sectional, self-reported data and variable recoding exist, findings in this study aligned with prior research on the decreased odds of severe depression among people with moderate alcohol consumption frequency, suggesting the validity of this study and meaningfulness of the findings. Future studies with utilization of longitudinal data are recommended for establishing cause-effect relationships between alcohol consumption and depression severity.

Reference

- Assanangkornchai, S., Nontarak, J., Aekplakorn, W., Chariyalertsak, S., Kessomboon, P., & Taneepanichskul, S. (2020). Socio-economic inequalities in the association between alcohol use disorder and depressive disorder among Thai adults: A population-based study. *BMC Psychiatry*, 20(1). <https://doi.org/10.1186/s12888-020-02958-6>
- Centers for Disease Control and Prevention. (n.d.). *About Nhis*. Centers for Disease Control and Prevention. <https://www.cdc.gov/nchs/nhis/about/index.html>
- Cho, Y., Lee, J. K., Kim, D. H., Park, J. H., Choi, M., Kim, H. J., Nam, M. J., Lee, K. U., Han, K., & Park, Y. G. (2019). Factors associated with quality of life in patients with depression: A nationwide population-based study. *PloS one*, 14(7), e0219455. <https://doi.org/10.1371/journal.pone.0219455>
- Hou, J. (2024). Association of Severity of Depressive Symptom and Weekly Alcohol Consumption Frequency after Considering Health Insurance, Sleep Hours and Smoking Status [BIST 0615 Final Project]. Rutgers University School of Public Health.
- Kim, S.-A., Kim, E., Morris, R. G., & Park, W.-S. (2015). Exploring the non-linear relationship between alcohol consumption and depression in an elderly population in Gangneung: The Gangneung Health Study. *Yonsei Medical Journal*, 56(2), 418. <https://doi.org/10.3349/ymj.2015.56.2.418>
- Kuehner, C. (2017). Why is depression more common among women than among men? *The Lancet Psychiatry*, 4(2), 146–158. [https://doi.org/10.1016/s2215-0366\(16\)30263-2](https://doi.org/10.1016/s2215-0366(16)30263-2)
- Lynn A. Blewett, Julia A. Rivera Drew, Miriam L. King, Kari C.W. Williams, Daniel Backman, Annie Chen, and Stephanie Richards. IPUMS Health Surveys: National Health Interview Survey, Version 7.4 [dataset]. Minneapolis, MN: IPUMS, 2024. <https://doi.org/10.18128/D070.V7.4>
- Monroe, S. M., & Harkness, K. L. (2022). Major depression and its recurrences: Life course matters. *Annual Review of Clinical Psychology*, 18(1), 329–357. <https://doi.org/10.1146/annurev-clinpsy-072220-021440>
- National Center for Health Statistics. (2010-2019). *National Health Interview Survey*. Hyattsville, MD: U.S. Department of Health and Human Services, Centers for Disease Control and Prevention.
- Qi, P., Huang, M., & Zhu, H. (2024). Association between alcohol drinking frequency and depression among adults in the United States: A cross-sectional study. *BMC Psychiatry*, 24(1). <https://doi.org/10.1186/s12888-024-06296-9>

Remes, O., Mendes, J. F., & Templeton, P. (2021). Biological, psychological, and social determinants of depression: A review of recent literature. *Brain Sciences*, *11*(12), 1633. <https://doi.org/10.3390/brainsci11121633>

Tu, Y., Qing, G., Chen, M., & Chen, H. (2024). Association between depression severity and trouble sleeping: A population-based study. *Medicine*, *103*(36). <https://doi.org/10.1097/md.00000000000039611>

U.S. Department of Health and Human Services. (n.d.). Depression. National Institute of Mental Health. <https://www.nimh.nih.gov/health/topics/depression>

Visontay, R., Mewton, L., Slade, T., Aris, I. M., & Sunderland, M. (2023). Moderate alcohol consumption and depression: A marginal structural model approach promoting causal inference. *American Journal of Psychiatry*, *180*(3), 209–217. <https://doi.org/10.1176/appi.ajp.22010043>

Wang, Z., & Pu, B. (2024). Joint effects of depression and social determinants of health on mortality risk among U.S. adults: A cohort study. *BMC Psychiatry*, *24*(1). <https://doi.org/10.1186/s12888-024-06159-3>